Project Charter (D1)

Analytics Capstone Project (ANCP), Term 5 · Module M1: Problem Framing

Project title	Next-Day Opening Price Prediction for NSE-Listed Stocks using End-of-Day Market Data and Company Fundamentals
Team members	Jayesh Santosh Shinde · Abhishek Manuel N · Abinash Pradhan
Faculty	Mr Nikhil D Jonathan
Date of submission	16 October 2026
Process model	CRISP-DM (Business Understanding → Data Understanding → Data Preparation → Modelling → Evaluation → Deployment)
Linked course outcomes	CO1 (problem formulation, success criteria, scope, data plan); CO6 (team plan, deadlines, ownership)

1. Business Context and Stakeholders

Every trading day on the National Stock Exchange (NSE) opens with a pre-open call auction (09:00–09:15 IST) in which the opening price is discovered. The opening price frequently differs from the previous day's close: news, quarterly results, global market moves and overnight sentiment all arrive while the market is shut. Traders, portfolio managers and treasury desks must decide *before* the open whether to place, hold or re-price orders, and today they mostly rely on the simple assumption that "tomorrow opens where today closed" or on informal judgement.

Stakeholder	Interest / decision	What they need from us
Retail and proprietary traders	Whether to place pre-open orders, and at what limit price	A predicted opening price per stock with a realistic range
Portfolio managers / fund dealers	Timing of planned buys and sells; avoiding a poor opening fill	Early warning of large expected gaps, ranked by stock
Risk managers	Overnight exposure and margin planning	Expected gap size and uncertainty band
Faculty / course assessors	Rigour, leakage-free validation, honest baselines	Documented, reproducible pipeline and evaluation

2. Problem Statement

> **Decision:** Before the NSE opens on day *t+1*, how should a trader price pre-open orders for a given stock?
> **Analytics question:** Using only information available at the close of day *t* (that day's OHLCV bar, recent price history, market-wide movement and the latest published company fundamentals), can we predict the opening price of each stock on day *t+1* more accurately than the naive "open = previous close" rule, and give a reliable range around that prediction?

Problem type: supervised regression on panel time-series data. The model predicts the *overnight gap*, $(\text{Open}_{t+1} \div \text{Close}_t) - 1$, which is then converted to a rupee opening price.

Predicting the gap rather than the raw price keeps the target comparable across stocks of very different price levels and avoids a model that merely copies yesterday's close.

Scope

In scope	Out of scope
• A universe of 5 large, liquid NSE stocks (RELIANCE, TCS, HDFCBANK, INFY, ICICIBANK), expanding to 20–50 NIFTY 50 constituents once the pipeline is stable • Daily data from January 2015 to the present • Price, volume and market-wide features plus quarterly fundamentals • One prediction per stock per trading day, with a 10–90% range • A batch pipeline and a Streamlit dashboard as the prototype	• Intraday or tick-level prediction, and prices after the open • Automated order placement or live trading • Investment advice or a profitability guarantee • Small-cap / illiquid stocks and derivatives (F&O) • News or social-media text sentiment (a possible extension only)

3. Success Metrics and Acceptance Criteria

All metrics are measured out of sample with walk-forward (expanding-window) validation over at least 5 time-ordered folds. The benchmark is the naive baseline $Open_{t+1} = Close_t$; a second baseline uses the historical mean gap from the training window.

Metric	Definition	Acceptance criterion
Mean Absolute Error (₹)	Average absolute difference between predicted and actual opening price	Lower than the naive baseline on average across folds, and in at least 4 of 5 folds
MAPE (%)	Mean absolute error as a percentage of the actual open	At least 5% relative improvement over the naive baseline (e.g. 0.60% → 0.57% or better)
Direction accuracy (%)	Share of days on which the sign of the predicted gap (up/down) is correct, excluding flat opens	Above 55%, and significantly above 50% (binomial test, $p < 0.05$)
Range coverage	Share of actual opens that fall inside the predicted 10–90% band	Between 75% and 85% (target 80%)
Statistical significance	Diebold–Mariano test of model errors vs baseline errors	Model better than baseline at the 5% level
Operational	End-to-end pipeline run after 16:00 IST	Predictions for all stocks ready before 09:00 IST next day; fully reproducible from the repository

Note: the naive baseline is hard to beat because most overnight gaps are small and close to random. If the model does not clear these thresholds, the team will report this honestly, together with an analysis of where (which stocks, which market conditions) any predictive signal does or does not exist.

4. Data Plan

Data	Source and access	Fields	Frequency, history and notes
Daily stock prices (OHLCV)	TejHQ market data API (`/v1/ohlcv/nse/{symbol}`); free tier, 1,000 requests/day	Date, open, high, low, close, previous close, volume, turnover, number of trades	Daily end of day, 2015 to present; pulled after 16:00 IST; cross-checked against NSE bhavcopy
Company fundamentals	CSV compiled from annual and quarterly reports / Screener.in now; Bloomberg terminal (institutional licence) when access is granted. Tracxn to be evaluated, though its coverage is mainly unlisted firms	Market cap (₹ cr), P/E, P/B, EPS (TTM), ROE %, debt-to-equity, sector	Quarterly; each row dated by its *results announcement date* so it is used only after it became public
Market context	Derived from the stock universe; NIFTY 50 index from the same API	Market-wide daily return and gap	Daily. Candidate extensions: overnight US market return, GIFT Nifty, INR/USD
Calendar	NSE holiday list	Trading days, day of week, holidays	Annual

Features (all known at day *t*'s close)

1-, 5- and 20-day returns; intraday return; today's gap; high–low range; position of close within the day's range; volume relative to its 20-day average; 20-day volatility; distance from 20-day moving average; day of week; market-wide return and gap; and the latest fundamentals (log market cap, P/E, P/B, EPS, ROE, debt-to-equity).

Preparation, quality and leakage controls

- **Time-aware validation only:** walk-forward splits; no random shuffling; at least 250 trading days of training history before the first test fold.
- **Point-in-time fundamentals:** backward-only as-of join on the announcement date, so no quarter's numbers reach days before publication.
- **Corporate actions:** days with overnight moves above ±20% (splits, bonuses, data errors) are flagged and excluded or adjusted.
- **Missing data and holidays:** gaps checked against the NSE calendar; missing fundamentals handled natively by the model (gradient boosting) rather than imputed with future values.
- **Bias:** survivorship bias acknowledged (current large caps only); large-cap results may not generalise to small caps.
- **Versioning:** raw and processed data stored by date in the team repository with a datasheet (D2).

Modelling approach (preview of D4)

Baselines first (open = previous close; historical mean gap), then linear regression with regularisation, then gradient-boosted trees (scikit-learn HistGradientBoosting, absolute-error loss). The uncertainty band comes from out-of-sample residual quantiles. Experiments are logged with parameters and fold-wise results.

Responsible data handling

The project uses only public market and company-level data; no personal data is collected, so the DPDP Act, 2023 consent requirements are not triggered. Data is used within each provider's licence terms (Bloomberg data stays within the institutional licence and is not redistributed). API keys are kept in an environment variable and never stored in code, documents or the repository. Outputs are presented as an analytical exercise, not investment advice.

Feasibility

Feasibility is high. The data is public and affordable, and a working prototype pipeline (data fetch, feature build, walk-forward training, prediction and Streamlit dashboard) already exists. It has so far been tested end to end on synthetic data only, so performance on real market data is still to be established.

5. Timeline and Deliverable Ownership

Each deliverable has one named lead owner, who is accountable for its quality and on-time submission; the other two members contribute. Every member leads three deliverables (the course requires at least two). Dates are team targets and will be aligned to the deadlines published by faculty.

Code	Deliverable	Module	Target date	Lead owner	Key activities
D1	Project charter	M1	16 Oct 2026	Jayesh Santosh Shinde	Problem framing, metrics, data plan, plan and roles
D2	Data audit report, datasheet, cleaned versioned dataset	M2	30 Oct 2026	Abhishek Manuel N	Pull 2015–present OHLCV, build fundamentals CSV, quality checks, corporate actions, datasheet
D3	EDA notebook and two-page insight summary	M3	9 Nov 2026	Abinash Pradhan	Gap distributions, volatility regimes, results-day effects, stock and sector segments, bias checks
D4	Modelling checkpoint with experiment log	M4	23 Nov 2026	Jayesh Santosh Shinde	Baselines, linear and boosted models, tuning, experiment tracking
D5	Evaluation memo	M5	2 Dec 2026	Abhishek Manuel N	Walk-forward design, leakage checklist, metrics vs baseline, significance tests, error analysis
D6	Deployment and monitoring plan (prototype)	M6	9 Dec 2026	Abinash Pradhan	Daily batch pipeline, Streamlit dashboard, drift monitoring, retraining schedule
D7	Draft report, model card, dry-run presentation	M7	18 Dec 2026	Jayesh Santosh Shinde	Report draft, model card and limitations, rehearsal and peer critique
E1	Final project report	End of term	As published	Abhishek Manuel N	Final report for a decision-maker audience

E2	Final presentation with individual Q&A	End of term	As published	Abinash Pradhan	Storyline, visuals; every member presents a section

Team roles and ways of working

Member	Primary role	Leads
Jayesh Santosh Shinde	Project coordination and modelling	D1, D4, D7
Abhishek Manuel N	Data engineering and evaluation	D2, D5, E1
Abinash Pradhan	Analysis, visualisation and deployment	D3, D6, E2

- Weekly team meeting with a written meeting log (attendance, decisions, actions, owners).
- All code and data versions in a shared team repository; contribution evidenced through commit history.
- A written agenda of blockers brought to each scheduled faculty consultation.
- Internal deadline two days before each published deadline for peer review by the non-lead members.

6. Key Risks and Mitigations

Risk	Impact	Mitigation
Model fails to beat the naive baseline (gaps are largely random)	High	Report honestly; analyse segments (results days, high-volatility periods) where signal exists; test exogenous global cues
API outage, rate limits or network restrictions	Medium	Cache all downloaded data locally; incremental top-ups; NSE bhavcopy as fallback source
Bloomberg / Tracxn access delayed	Medium	Fundamentals already handled through a CSV interface built from public reports; swap source later without code changes
Data leakage (look-ahead in features or fundamentals)	High	Point-in-time joins, walk-forward splits, leakage checklist reviewed in D5
Corporate actions distort prices	Medium	Flag and exclude extreme gaps; use adjusted prices where available
Uneven workload or missed deadlines	Medium	Named owners, internal deadlines, weekly log reviewed by the team

7. Sign-off

Name	Role	Signature / Date
Jayesh Santosh Shinde	Team member	
Abhishek Manuel N	Team member	
Abinash Pradhan	Team member	
Mr Nikhil D Jonathan	Faculty	